

# Requirements and Use Case Specification

Student Club Event Ticketing and Budget Portal

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This is the revised requirements record for the individual project. It keeps the five functional and two non-functional requirement IDs from Lab 1. The original submission is retained without edits in the originals folder.

## Functional requirements

| ID     | Type and priority | Description                                                                                                                                                                                                                                                    | Acceptance criteria                                                                                                                                                                                                                                                          | Rationale                                                                                                                   |
|--------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| FR-001 | Functional High   | The system shall route submitted budget requests through Faculty Coordinator, Finance Officer and Dean in that order. Each decision shall record the authenticated reviewer, stage, decision and time. Rejection shall stop the workflow and require a reason. | Pass: all three approvals occur in order and appear in the audit record. An early Dean approval, repeated sign-off or approval after rejection is refused. Fail: any stage is skipped or an unauthorised decision changes the status.                                        | This gives each office a chance to check a request before funds are committed.                                              |
| FR-002 | Functional High   | The system shall let a Club Lead create and edit their own draft proposal with an event name, date, venue, capacity and itemised budget. Submission shall validate required fields and the club allocation, then reserve the requested amount.                 | Pass: complete, affordable proposals enter Faculty pending. Missing fields, past dates, invalid amounts and requests above the remaining allocation stay out of the queue. The lead can reduce a draft budget and try again. Fail: invalid data or overspending is accepted. | Reviewers need complete event details. Reserving submitted budgets prevents two pending requests from using the same money. |
| FR-003 | Functional High   | The system shall issue a unique, signed QR ticket when a student registers for an approved, unexpired event with space available. The ticket shall identify the event and attendee. Repeat registration shall return the existing ticket.                      | Pass: different attendees receive different tickets; the same attendee receives one ticket per event. Draft, rejected, expired and full events reject registration. Fail: duplicate tickets or registrations above capacity are created.                                     | Each registration needs a verifiable ticket, and the number of attendees must stay within the venue capacity.               |

## Functional requirements continued

| ID     | Type and priority | Description                                                                                                                                                                                                                                        | Acceptance criteria                                                                                                                                                                                                                           | Rationale                                                                            |
|--------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| FR-004 | Functional High   | The system shall allow check-in staff to validate ticket text read from a QR code, accept an unused valid ticket once and mark it used. It shall reject altered, malformed, expired, wrong-event and previously used tickets.                      | Pass: the first scan is accepted and a second is rejected. Concurrent scans of the same ticket produce exactly one acceptance. Invalid input returns a rejection without a server error. Fail: an invalid ticket or repeat entry is accepted. | One attendee should not be able to share a ticket to admit several people.           |
| FR-005 | Functional Medium | The system shall show a Club Lead the current approval status, registration count and remaining allocation for their own events. The dashboard shall refresh status and counts within 5 seconds of a committed change during normal connected use. | Pass: an approval or registration is reflected on the open dashboard within 5 seconds; another lead cannot view the event. Fail: the display remains stale or exposes another lead's proposal.                                                | The lead should be able to follow an event without contacting each approving office. |

## Clarifications used in the prototype

A digital sign-off means a recorded decision by an authenticated reviewer, not a certificate-based digital signature. Money is stored as integer paise. Pending and approved requests reserve allocation; rejection releases it. The prototype has one configured allocation period and does not automatically renew annual budgets.

Students register themselves. Campus Admin is separated into three reviewer roles and check-in staff so that one ordinary account cannot skip approval stages. These roles make the original requirements explicit; they do not add payment collection or university account integration.

## Non-functional requirements

| ID      | Type and priority                   | Description                                                                                                                                                                                        | Acceptance criteria                                                                                                                                                                                                                                                                                             | Rationale                                                                                   |
|---------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| NFR-001 | Performance and security<br>High    | The check-in endpoint shall validate entry authenticity in under 100 ms under the agreed peak load. It shall require staff authorisation, verify the ticket signature and prevent duplicate entry. | Measure the complete HTTP validation request on the target server; use 20 simultaneous check-in clients as the initial peak-load assumption. Every request must stay below 100 ms, and tampering, role and replay tests must pass. Local results are reported separately from a production acceptance decision. | Slow validation creates queues, while weak validation allows forged or reused tickets.      |
| NFR-002 | Scalability and usability<br>Medium | The portal shall support at least 500 concurrent users during registration periods, with page loads under 2 seconds, and remain usable on mobile browsers.                                         | Test 500 concurrent authenticated users and measure browser page-load time below 2 seconds. At a 390-pixel viewport, forms must remain readable and usable; wide tables may scroll within their container. A server-only load test does not by itself prove browser-load performance.                           | Registration traffic can rise quickly after an announcement, and students often use phones. |

The performance values are acceptance targets, not claims about an unmeasured deployment. The local test method and observed results are in 6-Testing. Hardware scanner tests, real mobile-device tests and production load acceptance are still separate checks.

## **Use case UC01 Request budget allocation**

Primary actor: Club Lead. Supporting actor: Faculty Coordinator, who receives the request. Related requirements: FR-001, FR-002 and FR-005.

Preconditions: the Club Lead is logged in, belongs to a club and has saved a complete draft proposal for today or a future date. The club has a configured allocation.

Postconditions on success: the request is Faculty pending, its amount is reserved against the club allocation, and the lead can see the new status. On failure: the proposal stays a draft and no money is reserved.

### **Main success scenario**

1. The Club Lead opens a draft proposal and checks the event details and budget items.
2. The Club Lead selects Request budget allocation.
3. The system confirms ownership, draft status and the event date.
4. In one transaction, the system checks that the request fits within the remaining allocation.
5. The system changes the status to Faculty pending and reserves the amount.
6. The system displays confirmation. The Faculty Coordinator can now review the request.

### **Alternate flow A1 Budget exceeds the allocation**

At step 4, the request exceeds the remaining allocation. The system shows the remaining amount and leaves the proposal as a draft. The lead selects Edit draft, reduces the budget and saves it, then returns to step 2. If the lead leaves the page, the unchanged draft remains available.